



Muhammad Shehroz

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● ABOUT MYSELF

I am a motivated and ambitious Metallurgical Engineer. A recent graduate from NED University of Engineering and Technology. I am considerably solidly grounded in materials science, process optimization, and manufacturing techniques. During my undergraduate studies, I discovered interest in applying engineering principles to provide solutions for real-world problems and enhance industrial processes. I have developed the ability to solve problems analytically and efficiently through hands-on experience in material testing, quality control, and data analysis. I look forward to contributing my skills in metallurgy and engineering toward the success of a forward-thinking company.

● WORK EXPERIENCE

AMRELI STEELS LIMITED – DHABEJI, PAKISTAN

Business or Sector Manufacturing | **Department** SMS Production

TRAINEE ENGINEER – 17/02/2025 – CURRENT

Actively involved in induction furnace floor operations, gaining practical understanding of steel manufacturing processes.

Observed and studied the operation of continuous casting machines, gaining knowledge of the process flow, key equipment components, and quality parameters.

Supported mechanical and electrical maintenance activities, contributing to operational efficiency.

Participated in ladle operations and preparation, ensuring adherence to quality standards.

Gained exposure to refractories and their application in steel making.

Gained hands-on experience on SAP H4/HANA for material reservation.

HINOPAK MOTORS LIMITED

Business or Sector Manufacturing | **Department** Engineering and Development

INTERNEE – 02/03/2023 – 30/03/2023

Observed and learned about the processes involved in prototype development for automobile applications.

Gained experience in using SolidEdge software for drawing automobile parts.

Observed and analyzed Fiber Reinforced Plastic (FRP) processes.

Assisted in vehicle inspections, learning key quality control points.

PEOPLES STEEL MILLS LIMITED

Business or Sector Manufacturing | **Department** Quality Assurance Department

INTERNEE – 06/02/2024 – 24/02/2024

Supported Quality Assurance activities.

Studied and analyzed advanced steelmaking processes, including an introduction to rolling mill operations and quality parameters.

Assisted in and gained knowledge of destructive and non-destructive testing methods for final products, with exposure to testing rolled steel products.

● EDUCATION AND TRAINING

28/10/2020 – 02/08/2024

BACHELORS IN METALLURGICAL ENGINEERING NED University of Engineering and Technology

Field of study Metallurgical Engineering | **Final grade** 2.9 CGPA

Field of study Engineering | **Final grade** B**Field of study** Computer Science | **Final grade** A

● LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	A1	C2	A2	A2	C1
CHINESE	A1	A1	A1	A1	A1

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user

● SKILLS

Microsoft Office package: Microsoft Word, Excel, PowerPoint, Access | Microsft Project | Origin software for data analysis | Echem analyst Gamry | SAP

● PROJECTS

09/10/2023 – 02/08/2024

Experimental and Simulation test of projectile impact on high strength steels for Automobile Applications.

Conducted mechanical testing such as tensile and hardness test on Ramor 500/550 to obtain simulation input parameters.

Simulated projectile impact on Ramor 500/550 steels for automotive safety, using mechanical testing data

Developed explicit dynamic finite element models in Ansys AutoDyn to simulate impact on Ramor 500/550.

Implemented Johnson Cook material model calibrated with Ramor 500/550 test data.

Analyzed and compared simulation results with mechanical testing behavior of Ramor 500/550.

11/03/2023 – 17/04/2023

Development of thin films for packaging applications.

Developed thin films using Poly vinyl Alcohol (PVA) and Toluene.

Performed Thermo Gravimetric Analysis (TGA) and Differnetial Scanning Calorimetry (DSC) to study their properties.

● CERTIFICATIONS

08/09/2024 – 25/09/2024

ISO 9001-2015- Quality Management System

Gained knowledge about the ISO Quality Management System.

Learned about the Quality tools for Quality Assurance.

Gained knowledge about the customer satisfaction and other key aspects of Quality Management System